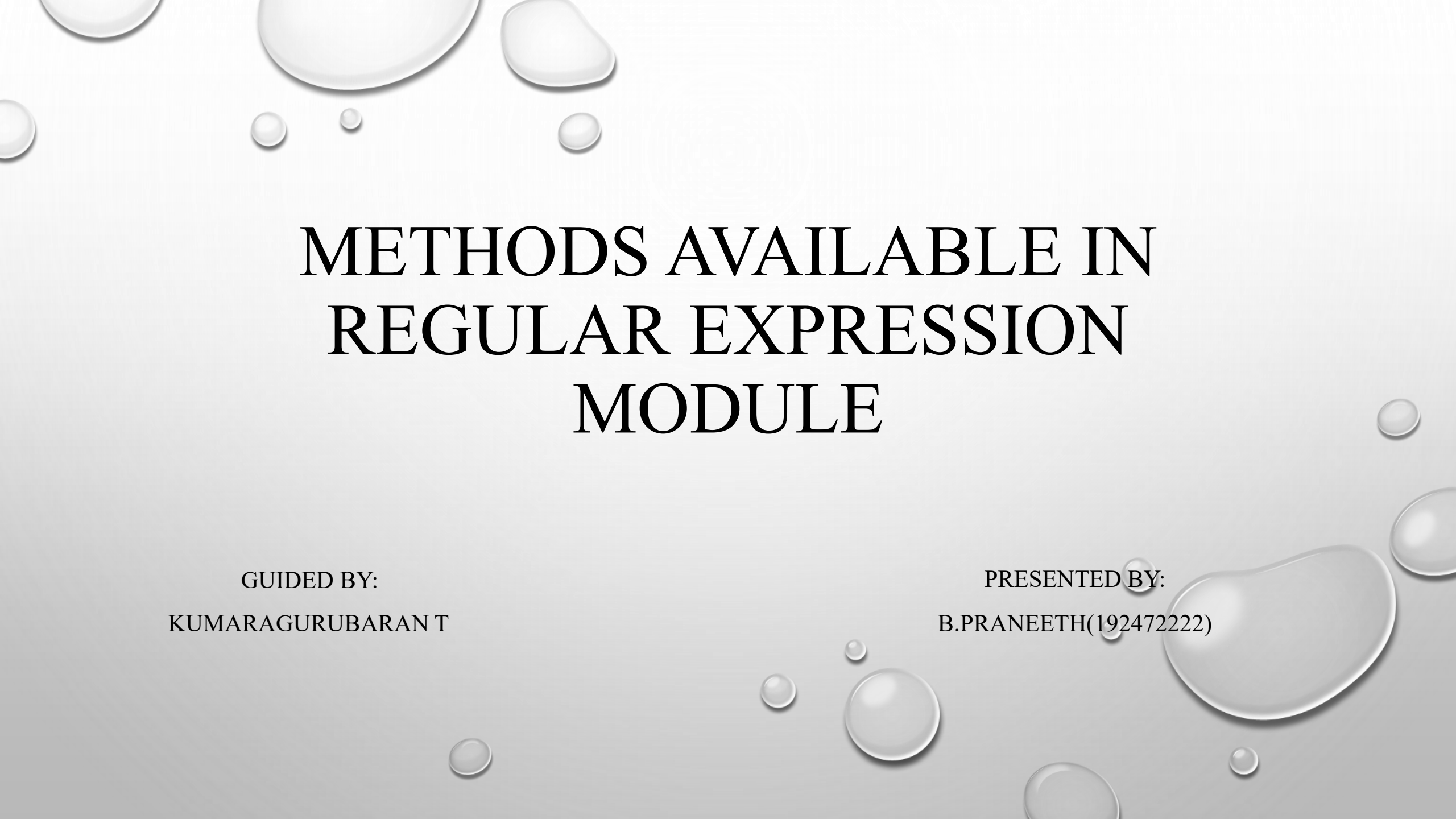


The background of the slide is a light gray gradient, decorated with numerous realistic water droplets of various sizes. Some droplets are large and prominent, while others are small and subtle, scattered across the top and bottom edges of the slide.

METHODS AVAILABLE IN REGULAR EXPRESSION MODULE

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INTRODUCTION

- RE stands for regular expressions.
- Regular expression (regex) is a sequence of characters that defines a search pattern.
- In python, regular expressions are supported through the built-in RE module.
- Regex is used to search, match, extract, validate, and replace text efficiently.
- It helps process large amounts of text with simple pattern-based rules.

WHY USE REGULAR EXPRESSIONS ?

- **Efficient text searching** – quickly finds specific words or patterns in text.
- **Data validation** – validates inputs such as email addresses, phone numbers, passwords, and pin codes.
- **Text extraction** – extracts useful information like dates, numbers, urls, and email addresses from text.
- **Search and replace** – easily replaces words or patterns in a string.
- **Text splitting** – splits text based on a specified pattern instead of a fixed character.

APPLICATIONS

- **Email validation** – checks whether an email address is in the correct format.
- **Phone number validation** – verifies valid mobile or telephone numbers.
- **Password validation** – ensures passwords meet security requirements.
- **Data extraction** – extracts emails, urls, dates, numbers, and other information from text.
- **Search and replace** – finds and replaces words or patterns in documents and files.

ADVANTAGES

- **Fast and efficient** – performs text searching and pattern matching quickly.
- **Saves time** – reduces the amount of code needed for string processing.
- **Powerful pattern matching** – can match simple or complex text patterns.
- **Improves code readability** – makes text-processing tasks more concise and easier to maintain.
- **Supports data validation** – easily validates emails, phone numbers, passwords, and other inputs.

OBJECTIVES

- To understand the concept of regular expressions (regex) in python.
- To learn how the re module is used for pattern matching.
- To search for specific text within strings efficiently.
- To extract useful information from text data.
- To replace or modify text using regular expressions.

LIMITATIONS

- **Difficult to read** – complex regex patterns can be hard to understand.
- **Hard to maintain** – long expressions become difficult to modify and debug.
- **Steep learning curve** – beginners may find regex syntax confusing.
- **Not suitable for complex logic** – cannot easily handle deeply nested or context-dependent structures.
- **Performance issues** – very complex patterns may slow down execution on large text.

CONCLUSION

- The python re module provides powerful support for working with regular expressions (regex).
- It helps perform efficient pattern matching, searching, extracting, splitting, and replacing text.
- The re module is widely used for input validation, data cleaning, log analysis, web scraping, and text processing.
- Learning the re module enables developers to write shorter, more efficient, and more reliable python programs for handling textual data.

The image features a light gray background with a subtle radial gradient. In the top-left and bottom-right corners, there are several realistic water droplets of varying sizes, rendered with soft shadows and highlights to give them a three-dimensional appearance. Centered in the upper half of the image is a faint, circular, embossed-style seal or logo, which appears to be a crest or emblem, though its details are too light to discern clearly.

THANK YOU

GPS PHOTO

